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Fundamental Concepts of Data Security

W01 - Information Security Concepts

Le Hai Duong, PhD. (lhduong@hcmiu.edu.vn)

Course Info.

Materials

- **Textbook:**
  - [Stallings2022] Stallings, W., Brown, L. (2022). *Cryptopraphy and Network Security* (8th ed.). Upper Saddle River, NJ, USA: Pearson Education.
- **Extra reading:**
  - [Du2017] Du, W. (2017). *Computer Security: a hands-on approach*. CreateSpace.

Tentative Schedule

Session	Title	Reading
	SECURITY OVERVIEW	
1	Information Security Concepts	Chapter 1
	SYMMETRIC CIPHERS	
2	Classical Encryption Techniques	Chapter 3
3	Block Cipher and the Data Encryption Standard (DES)	Chapter 4
4	Advanced Encryption Cipher (AES)	Chapter 6
5	Block Cipher Operation	Chapter 7
6	Random Bit Generation and Stream Ciphers	Chapter 8
7	**Review**	
MIDTERM		
	ASYMMETRIC CIPHERS	
8	Public-key Cryptography and RSA	Chapter 9
	CRYPTOGRAPHIC DATA INTEGRITY ALGORITHMS	
9	Cryptographic Hash Functions	Chapter 11
10	Message Authentication Codes	Chapter 12
11	Digital Signatures	Chapter 13
	MUTUAL TRUST	
12	Cryptographic Key Management and Distribution	Chapter 15
	NETWORK AND INTERNET SECURITY	
13	Transport Layer Security	Chapter 17
14	IP Security and Cloud Security	Chapters 20, 22
15	**Review**	
FINAL		

Gradings

- Labs, homework, quizzes 30%
- Midterm exam 30%
- Final exam 40%

Outline

- Cybersecurity and Information Security
- Security Attacks
- Security Services
- Security Mechanisms
- Cryptography
- Cryptography in Practice

Three fundamental questions

- What "**things**" do need to protect?
- How are those "**things**" threatened?
- What can we do to counter those threats?

Cybersecurity and Information Security

Cybersecurity

- Protect stored, transmitted, and processed information in
  - networked systems
  - digital devices
  - network devices and transmission lines

Information Security

- Ensure confidentiality, integrity and availability of information —> CIA Triad
- Ensure authenticity, accountability, nonrepudiation, and reliability

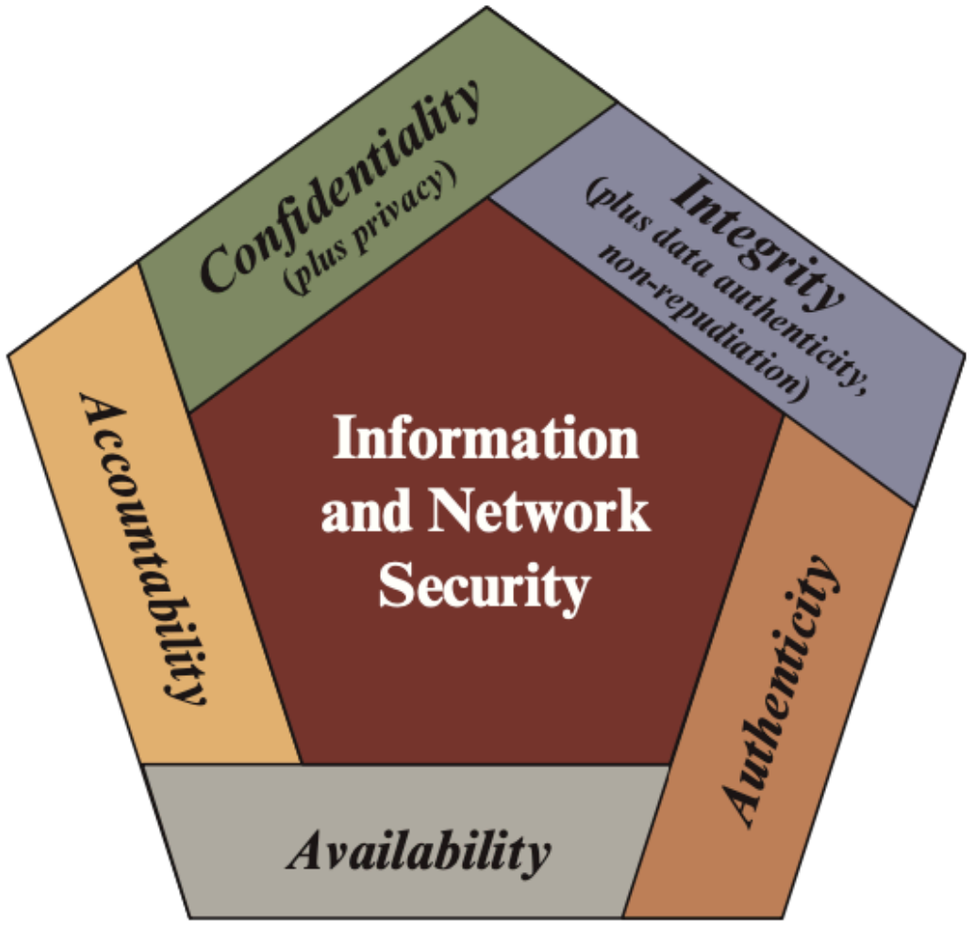


Figure 1.1 Essential Information and Network Security Objectives

CIA Triad

Primary goals and objectives of a security infrastructure

- **Confidentiality**: Preserving authorized restrictions on information access and disclosure, including means for protecting personal privacy and proprietary information. A loss of confidentiality is the unauthorized disclosure of information.
    - Data confidentiality
    - Privacy
- **Integrity:** Guarding against improper information modification or destruction, including ensuring information nonrepudiation and authenticity. A loss of integrity is the unauthorized modification or destruction of information.
    - Data integrity
    - System integrity
- **Availability:** Ensuring timely and reliable access to and use of information. A loss of availability is the disruption of access to or use of information or an information system.

DAD Triad

Failures of security protections in the CIA Triad

- **Disclosure** occurs when sensitive or confidential material is accessed by unauthorized entities, it is a violation of confidentiality
- **Alternation** occurs when data is either maliciously or accidentally changed, it is a violation of integrity.
- **Destruction** occurs when a resource is damaged or made inaccessible to authorized users (technically we usually call the later denial of service (DoS)), it is a violation of availability.

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AAA services

Core security mechanism of all security environments

Five processes of security:

- **Identification** is claiming to be an identity when attempting to access a secured area or system.
- **Authentication** is proving that you are that claimed identity.
- **Authorization** is defining the permissions (i.e., allow/grant and/or deny) of a resource and object access for a specific identity or subject.
- **Auditing** is recording a log of the events and activities related to the system and subjects.
- **Accounting** (aka *accountability*) is reviewing log files to check for compliance and violations in order to hold subjects accountable for their actions, especially violations of organizational security policy.

Security Attacks

Computer Security Terminology (RFC 2828)

- **System Resource (Asset)**: Data contained in an information system; or a service provided by a system; or a system capability, such as processing power or communication bandwidth; or an item of system equipment (i.e., a system component-hardware, firmware, software, or documentation); or a facility that houses system operations and equipment.
- **Adversary**: An entity that attacks, or is a threat to a system. **Security violation**: An act or event that disobeys or otherwise breaches security policy.
- **Security policy**: A set of rules and practices that specify or regulate how a system or organization provides security services to protect sensitive and critical system resources.
- **Security mechanisms**: A process (or a device incorporating such a process) that can be used in a system to implement a security service that is provided by or within the system.
- **Risk**: An expectation of loss expressed as the probability that a particular threat will exploit a particular vulnerability with a particular harmful result.

Risk = Threat * Vulnerability

Vulnerabilities, Threats and Attacks

- A **vulnerability** is simply a weakness that can be exploited by an attacker to perform unauthorized actions within a computer or network system.
    - corrupted (loss of integrity)
    - leaky (loss of confidentiality)
    - unavailable or very slow (loss of availability)
- **Threats**
    - capable of exploiting vulnerabilities
    - represent potential security harm to an asset
- **Attacks** (threats carried out)
    - **Passive** – attempt to learn or make use of information from the system that does not affect system resources
    - **Active** – attempt to alter system resources or affect their operation
    - **Insider** – initiated by an entity inside the security parameter
    - **Outsider** – initiated from outside the perimeter

Types of Attacks

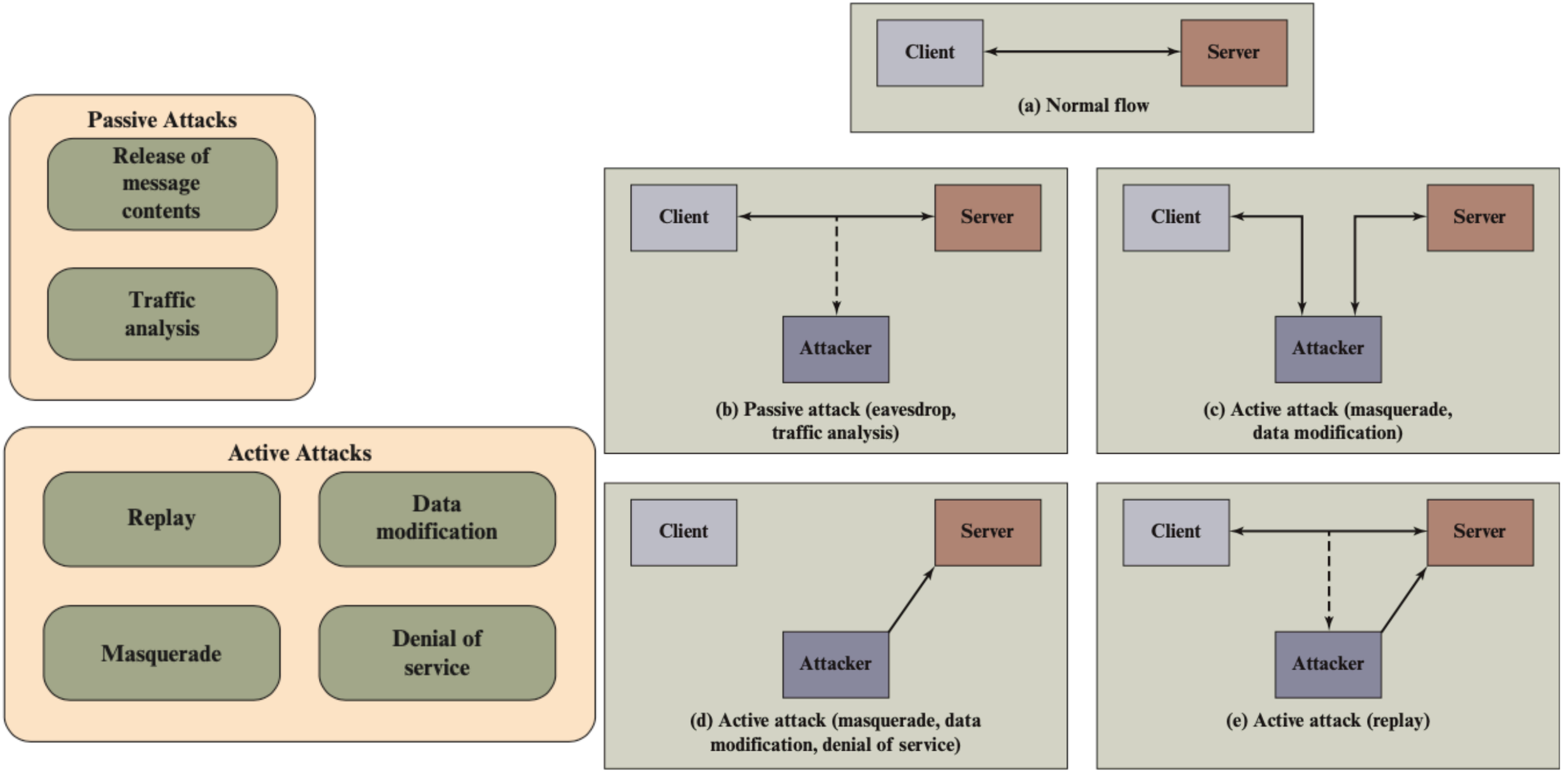


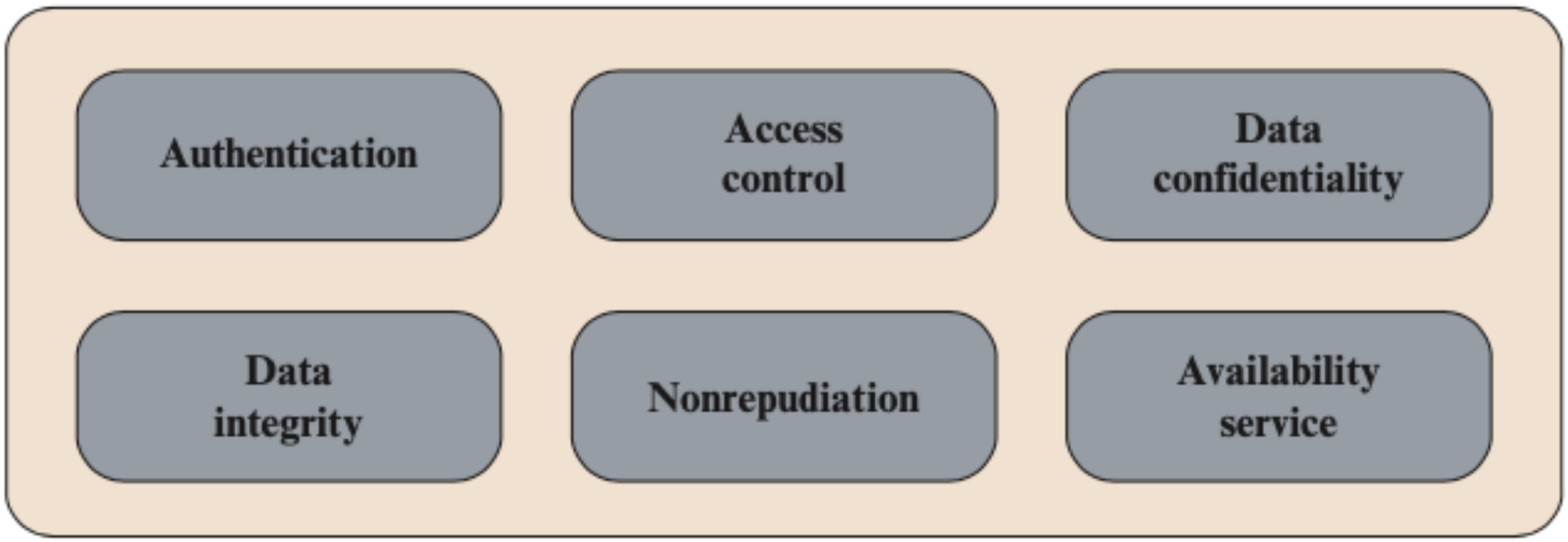
Figure 1.3 Security Attacks

Countermeasures

- Means used to deal with security attacks
    - Prevent
    - Detect
    - Recover
- Goal is to minimize residual level of risk to the assets
- May itself introduce new vulnerabilities
- Residual vulnerabilities may remain

Security Services

Security Services

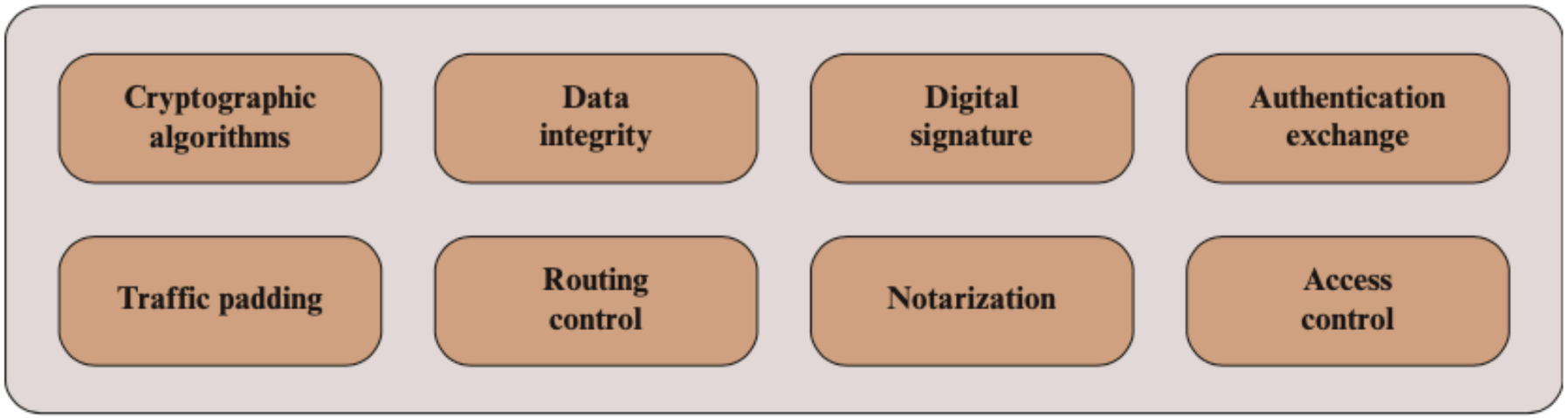


(b) Services

- **Authentication**: ensure a communication is authentic
    - Peer entity authentication
    - Data origin authentication
- **Access control**: ability to limit and control the access to host systems and applications via communications links
- **Data confidentiality**: protection of transmitted data from passive attacks
- **Data integrity**: make sure that received data are not altered
- **Nonrepuditation**: prevents either sender or receiver from denying a transmitted messages
- **Availability service**: protects a system to ensure its availability

Security Mechanisms

Security Mechanisms



(c) Mechanisms

- A process (or a device incorporating such a process) that can be used in a system to implement a security service that is provided by or within the system

Cryptography

Cryptography Algorithms

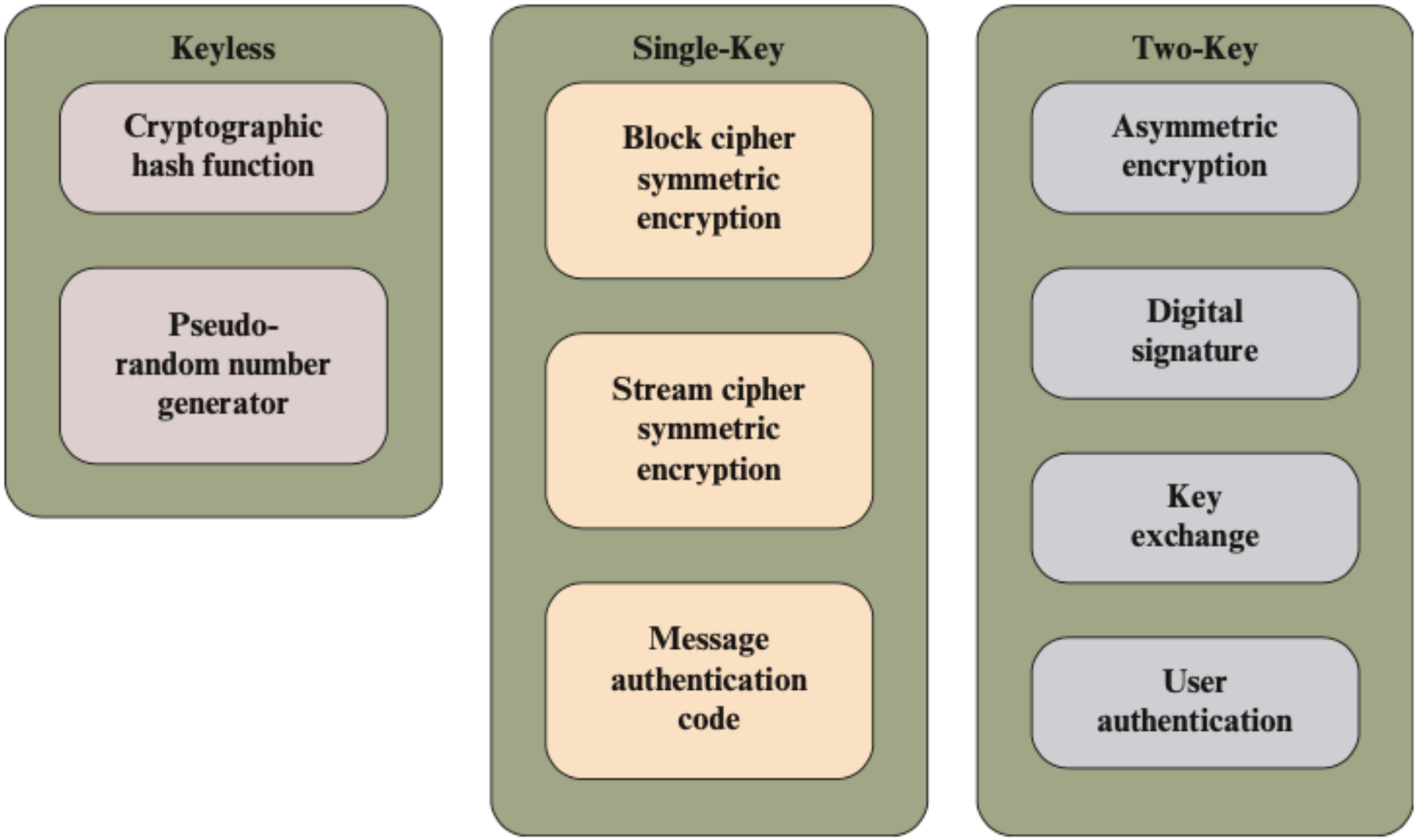


Figure 1.4 Cryptographic Algorithms

Cryptography in Practice

Secure Web Communications



How do you whether your username and password are protected while traversing the Internet?

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Overview

Main origin

Reload to view details

Security overview

This page is secure (valid HTTPS).

- Certificate - valid and trusted

  The connection to this site is using a valid, trusted server certificate issued by Sectigo RSA Domain Validation Secure Server CA.

  View certificate

- Connection - secure connection settings

  The connection to this site is encrypted and authenticated using TLS 1.2, ECDHE_RSA with X25519, and AES_128_GCM.

- Resources - all served securely

  All resources on this page are served securely.